



**UNITED INTERNATIONAL UNIVERSITY**  
 Department of Computer Science and Engineering (CSE)  
**Course Syllabus**

| 1   | Course Title                                                                                                                     | Artificial Intelligence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
|-----|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------------------|-----|----------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------------|
| 2   | Course Code                                                                                                                      | CSE 3811                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| 3   | Trimester and Year                                                                                                               | Fall 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| 4   | Pre-requisites                                                                                                                   | CSE 2217                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| 5   | Credit Hours                                                                                                                     | 3.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| 9   | Instructor's Name                                                                                                                | <a href="#">Nahid Ferdous Aurna</a> (Assistant Prof. CSE, UIU)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| 10  | Email                                                                                                                            | aurna@cse.uiu.ac.bd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| 11  | Office                                                                                                                           | Room #318(B)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| 12  | Counseling Hours                                                                                                                 | Saturday & Tuesday: 12:30 PM- 3:30 PM<br>Sunday & Wednesday: 11:00 AM- 01:00 PM,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| 13  | Text Book                                                                                                                        | Artificial Intelligence - A Modern Approach (3 <sup>rd</sup> edition) by Stuart Russel and Peter Norvig                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| 14  | Reference                                                                                                                        | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| 15  | Course Contents (approved by UGC)                                                                                                | Survey and concepts in Artificial Intelligence; Problem-solving agents; Uninformed and Informed search techniques; Local Search Techniques; Game playing; Constraint Satisfaction Problems; Bayesian learning; Supervised Learning: Classification, Perceptrons; Stationary processes and Markov assumptions; Hidden Markov Models; Human Aware AI Systems.                                                                                                                                                                                                                                                  |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| 16  | Course Outcomes (COs)                                                                                                            | <table><tr><th>COs</th><th>Description</th></tr><tr><td>CO1</td><td>Understand the role of decision-making strategies in simulating intelligence, and gain familiarity with approaches towards such.</td></tr><tr><td>CO2</td><td>Analyze and apply search strategies for solutions to problems with complete information.</td></tr><tr><td>CO3</td><td>Understand and apply reasoning strategies for inference in the presence of incomplete information.</td></tr><tr><td>CO4</td><td>Demonstrate familiarity with approaches to exploiting regularity in data and areas of application.</td></tr></table> | COs | Description       | CO1 | Understand the role of decision-making strategies in simulating intelligence, and gain familiarity with approaches towards such. | CO2        | Analyze and apply search strategies for solutions to problems with complete information. | CO3 | Understand and apply reasoning strategies for inference in the presence of incomplete information. | CO4 | Demonstrate familiarity with approaches to exploiting regularity in data and areas of application. |
| COs | Description                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| CO1 | Understand the role of decision-making strategies in simulating intelligence, and gain familiarity with approaches towards such. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| CO2 | Analyze and apply search strategies for solutions to problems with complete information.                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| CO3 | Understand and apply reasoning strategies for inference in the presence of incomplete information.                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| CO4 | Demonstrate familiarity with approaches to exploiting regularity in data and areas of application.                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| 17  | Teaching Methods                                                                                                                 | Lecture, Case Studies.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| 18  | CO with Assessment Methods                                                                                                       | <table><tr><th>CO</th><th>Assessment Method</th><th>(%)</th></tr><tr><td>-</td><td>Attendance</td><td>5</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | CO  | Assessment Method | (%) | -                                                                                                                                | Attendance | 5                                                                                        |     |                                                                                                    |     |                                                                                                    |
| CO  | Assessment Method                                                                                                                | (%)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |
| -   | Attendance                                                                                                                       | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |                   |     |                                                                                                                                  |            |                                                                                          |     |                                                                                                    |     |                                                                                                    |

|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|----|-------------------------------------|--|----------|--------------|--|--|--|-------------------------|--|--|--|--|--|
|    |                                     |  | -        | Assignments  |  |  |  | 5<br>(Best 1 out of 2)  |  |  |  |  |  |
|    |                                     |  | -        | Class Tests  |  |  |  | 20<br>(Best 3 out of 4) |  |  |  |  |  |
|    |                                     |  | CO1, CO2 | Midterm exam |  |  |  | 30                      |  |  |  |  |  |
|    |                                     |  | CO3, CO4 | Final exam   |  |  |  | 40                      |  |  |  |  |  |
| 19 | Mapping of COs and Program outcomes |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |
|    |                                     |  |          |              |  |  |  |                         |  |  |  |  |  |

|  |    |                                                                      |              |            |                     |  |
|--|----|----------------------------------------------------------------------|--------------|------------|---------------------|--|
|  |    |                                                                      | MIDTERM EXAM |            |                     |  |
|  | 13 | Basic probability notation; Inference using full joint distribution  | 3            | 13.2, 13.3 | Lecture             |  |
|  | 14 | Probabilistic independence; Bayes' rule and its use                  | 3            | 13.4, 13.5 | Lecture             |  |
|  | 15 | Semantics of Bayesian networks; Exact inference in Bayesian networks | 3            | 14.2, 14.4 | Lecture             |  |
|  | 16 | Exact inference in Bayesian networks;                                | 3            | 14.4       | Lecture             |  |
|  | 17 | Exact inference in Bayesian networks;                                | 3            | 14.4       | Lecture             |  |
|  | 18 | Naive Bayes Model                                                    | 3            | 20.2       | Class Test; Lecture |  |
|  | 19 | Stationary processes and Markov assumptions                          | 3            | 15.1       | Lecture             |  |
|  | 20 | Stationary processes and Markov assumptions; Hidden Markov models    | 3            | 15.1, 15.3 | Lecture             |  |
|  | 21 | Decision Tree Learning                                               | 4            | 18.3       | Lecture             |  |
|  | 22 | Decision Tree Learning                                               | 4            | 18.3       | Lecture             |  |
|  | 23 | Single Layer Perceptrons                                             | 4            | 18.7       | Class Test; Lecture |  |
|  | 24 | Multilayer Perceptrons                                               | 4            | 18.7       | Lecture             |  |

### **Appendix 1: Assessment Methods**

| Assessment Types | Marks           |
|------------------|-----------------|
| Attendance       | 5%              |
| Assignments      | 5%(1 out of 2)  |
| Class Tests      | 20%(2 out of 4) |
| Mid Term         | 30%             |
| Final Exam       | 40%             |

### **Appendix 2: Grading Policy**

| Letter Grade | Marks % | Grade Point | Letter Grade | Marks% | Grade Point |
|--------------|---------|-------------|--------------|--------|-------------|
| A (Plain)    | 90-100  | 4.00        | C+ (Plus)    | 70-73  | 2.33        |
| A- (Minus)   | 86-89   | 3.67        | C (Plain)    | 66-69  | 2.00        |
| B+ (Plus)    | 82-85   | 3.33        | C- (Minus)   | 62-65  | 1.67        |

|            |       |      |           |       |      |
|------------|-------|------|-----------|-------|------|
| B (Plain)  | 78-81 | 3.00 | D+ (Plus) | 58-61 | 1.33 |
| B- (Minus) | 74-77 | 2.67 | D (Plain) | 55-57 | 1.00 |
|            |       |      | F (Fail)  | <55   | 0.00 |

### **Appendix-3: Program outcomes**

| <b>P0s</b>  | <b>Program Outcomes</b>                                                                                                                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P01</b>  | An ability to apply knowledge of mathematics, science, and engineering                                                                                                                                                        |
| <b>P02</b>  | An ability to identify, formulate, and solve engineering problems                                                                                                                                                             |
| <b>P03</b>  | An ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability |
| <b>P04</b>  | An ability to design and conduct experiments, as well as to analyze and interpret data                                                                                                                                        |
| <b>P05</b>  | An ability to use the techniques, skills, and modern engineering tools necessary for engineering practice                                                                                                                     |
| <b>P06</b>  | The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context                                                                                    |
| <b>P07</b>  | A knowledge of contemporary issues                                                                                                                                                                                            |
| <b>P08</b>  | An understanding of professional and ethical responsibility                                                                                                                                                                   |
| <b>P09</b>  | An ability to function on multidisciplinary teams                                                                                                                                                                             |
| <b>P010</b> | An ability to communicate effectively                                                                                                                                                                                         |
| <b>P011</b> | Project Management and Finance                                                                                                                                                                                                |
| <b>P012</b> | A recognition of the need for, and an ability to engage in lifelong learning                                                                                                                                                  |